Sahaj Bhadja

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Summary

AI-native software engineer with 2+ years building operational tools, automation workflows, and lightweight applications. Strong in Python, JavaScript/TypeScript, SQL, prompting discipline, and AI-output verification.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant platform services on **Azure SQL** and Blob Storage for 500+ users, optimizing indexed queries and data-access patterns to reduce dashboard latency from 1.2s to 350ms in production.
- Built fault-tolerant execution workflows with explicit state transitions and failure isolation, processing 1K+ AI jobs while improving throughput 3x under concurrent load and documenting repeatable operational behaviors for handoffs.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Flask** services and **REST APIs** for AI grading workflows serving 10K+ daily requests, integrating secure file ingestion, scoring pipelines, and instructor-facing tools across customer-facing production software.
- Applied prompt iteration, model switching, and output validation across multimodal grading workflows, reducing feedback turnaround by 60% while supporting non-technical instructors through reliable AI-assisted operational tooling daily.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built and evaluated spatial-temporal planning pipelines over 100K+ trajectory logs, improving closed-loop decision stability by 15% while strengthening reproducibility, debugging, and interdisciplinary communication across fast-moving simulation workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered microservice-based software with **Jenkins**, **PyTest**, and **Mockito** automation across Windows, Linux, and macOS, improving release reliability through 20+ unit and integration test suites in production.
- Optimized large-scale API and ETL workflows with **AWS Glue**, **S3**, and object-oriented design patterns, improving response times by 30% and increasing pipeline efficiency by 68% for operational data flows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed business workflow tools with **REST APIs**, **WebSockets**, microservices, and **Jenkins CI/CD**, supporting 2M+ synchronized records while improving rollout speed, maintainability, stakeholder alignment, and operational consistency.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
AI-Assisted Building: Claude, Claude Code, Cursor, prompt design, prompt iteration, AI-output verification, model evaluation, governance awareness, tool documentation
Applications & Automation: Flask, FastAPI, Node.js, REST APIs, WebSockets, dashboards, lightweight applications, scripts, workflow automation, microservices
Data & Platforms: PostgreSQL, MySQL, SQL Server, Azure SQL, AWS RDS, S3, Blob Storage, ETL pipelines, data validation, query optimization, Linux
Engineering Practices: Git, Jenkins, Docker, CI/CD, debugging, unit testing, integration testing, stakeholder communication, knowledge transfer, Agile/Scrum

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Implemented consent-based patient-history access, Twilio appointment alerts, Sentry monitoring, and scalable TypeScript-backed workflows to reduce administrative workload by 30%, emphasizing healthcare operations, documentation, and business-friendly tool behavior.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management while creating developer-friendly workflows that help non-experts work productively with operational data.